

Summer Math Adventure

Weeks 6 – 8 Extension Booklet

Exponents · Expressions & Equations · Probability

3 Weeks · 15 Daily Pages · 3 Quizzes · Full Answer Key · ~45 min/day

Name: _____

Summer: 2025

- | | |
|--|--------------------------------------|
| ☐ Week 6: Exponents | ☐ Week 6 Quiz |
| ☐ Week 7: Expressions & Eq. | ☐ Week 7 Quiz |
| ☐ Week 8: Probability | ☐ Week 8 Quiz |

Tools needed: pencil, calculator (Week 6 challenge only)

Summer Math Adventure

W6
D1

Week 6 · Day 1 | What Is an Exponent?

Name: _____

Date: _____

EXPONENTS

Key Idea: An *exponent* tells you how many times to multiply the *base* by itself.

3^4 means $3 \times 3 \times 3 \times 3 = \mathbf{81}$ | base = 3, exponent = 4

Special names: $5^2 =$ "five squared" | $5^3 =$ "five cubed"

Remember: Any number to the power of 0 equals 1. ($7^0 = 1$)

Part A – Label the Parts

For each expression, write the base, the exponent, and read it aloud.

1. 2^5

Base:

Exponent:

Read as:

2. 10^3

Base:

Exponent:

Read as:

3. 6^2

Base:

Exponent:

Read as:

4. 4^0

Base:

Exponent:

Value:

Part B – Expand & Evaluate

Write as repeated multiplication, then find the value.

5. $3^3 = ___ \times ___ \times ___ =$

6. $2^6 = ___ \times ___ \times ___ \times ___ \times ___ \times ___ =$

7. $5^2 = ___ \times ___ =$

8. $10^4 = ___ \times ___ \times ___ \times ___ =$

Name: _____ Date: _____

EXPONENTS

Pattern: $10^1=10$ | $10^2=100$ | $10^3=1,000$ | $10^4=10,000$
Each time the exponent goes up by 1, you multiply by 10 (add a zero).
Shortcut: $10^n = 1$ followed by n zeros.

Part A – Complete the Powers of 10 Table

Exponential	Expanded	Standard Form	Number of Zeros
10^1	10	10	1
10^2	10×10		
10^3			
10^4			
10^5			
10^6			

Part B – Multiply Using Powers of 10

Use the pattern: $4 \times 10^3 = 4 \times 1,000 = 4,000$

1. $7 \times 10^2 =$

2. $3 \times 10^4 =$

3. $9 \times 10^1 =$

4. $6 \times 10^5 =$

5. $25 \times 10^3 =$

6. $14 \times 10^2 =$

Part C – Fill in the Exponent

Summer Math Adventure

W6
D3

Week 6 · Day 3 | Comparing Exponential Expressions

Name: _____

Date: _____

EXPONENTS

Strategy: Always evaluate each expression first, then compare.

Example: Is 2^5 or 5^2 greater?

$$2^5 = 32 \quad 5^2 = 25 \quad \text{So } 2^5 > 5^2$$

Part A – Evaluate Each Side, Then Compare (<, >, =)

1. 3^4 4^3
 $3^4 = \underline{\hspace{1cm}}$ $4^3 = \underline{\hspace{1cm}}$

2. 2^7 7^2
 $2^7 = \underline{\hspace{1cm}}$ $7^2 = \underline{\hspace{1cm}}$

3. 5^3 3^5
 $5^3 = \underline{\hspace{1cm}}$ $3^5 = \underline{\hspace{1cm}}$

4. 6^2 2^6
 $6^2 = \underline{\hspace{1cm}}$ $2^6 = \underline{\hspace{1cm}}$

5. 10^3 1,000

6. 4^2 2^4

Part B – Order from Least to Greatest

7. 2^3 3^2 2^4 4^1
Values: $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$
Order:

8. 5^0 2^5 3^3 10^2
Values: $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$
Order:

Part C – True or False? Correct the False Ones.

9. $2^3 = 6$ T / F
If false, correct:

10. $3^0 = 0$ T / F
If false, correct:

11. $10^3 = 1,000$ T / F

12. $1^{100} = 100$ T / F

Summer Math Adventure

W6
D4

Week 6 · Day 4 | Order of Operations with Exponents

Name: _____

Date: _____

EXPONENTS

PEMDAS Order:

1. Parentheses 2. Exponents 3. Multiplication / Division (left→right) 4. Addition / Subtraction (left→right)

Example: $2 + 3^2 \times 4 = 2 + 9 \times 4 = 2 + 36 = 38$

Part A – Evaluate Step by Step

Show each step. Circle the operation you do first.

1. $5 + 2^3$

Step 1: ____

Answer:

2. $4^2 - 10$

Step 1: ____

Answer:

3. 3×2^4

Step 1: ____

Answer:

4. $(3 + 1)^2$

Step 1: ____

Answer:

5. $2^3 + 3^2$

Step 1: ____ Step 2: ____

Answer:

6. $50 - 2^2 \times 3$

Step 1: ____ Step 2: ____ Step 3: ____

Answer:

Part B – Multi-Step Expressions

7. $(2 + 3)^2 - 4 \times 3 =$

8. $6^2 \div (3 + 1) + 5 =$

Week 6 · Day 4 – PEMDAS with Exponents

Score: _____ / 11

Summer Math Adventure

Week 6 · Day 5 | Perfect Squares & Square Roots

Name: _____ Date: _____

EXPONENTS

Perfect Square: A number that equals an integer times itself. $25 = 5 \times 5 = 5^2$

Square Root ($\sqrt{}$): The opposite — what number times itself gives you this? $\sqrt{25} = 5$

First 12 perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Part A – Perfect Squares Chart

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1											

Part B – Square Roots

1. $\sqrt{36} =$

2. $\sqrt{81} =$

3. $\sqrt{144} =$
- _____

4. $\sqrt{49} =$

5. $\sqrt{121} =$

6. $\sqrt{1} =$
- _____

Part C – Is It a Perfect Square?

Circle YES or NO. If yes, write its square root.

7. 50 YES / NO $\sqrt{} = \underline{\hspace{1cm}}$

8. 64 YES / NO $\sqrt{} = \underline{\hspace{1cm}}$
9. 100 YES / NO $\sqrt{} = \underline{\hspace{1cm}}$

10. 72 YES / NO $\sqrt{} = \underline{\hspace{1cm}}$

Part D – Area & Square Roots in Context

11. A square garden has an area of 64 m². What is the length of one side?

Side = _____
12. A checkerboard has 64 small squares arranged in a perfect square pattern. How many rows does it have?

Rows = _____

Summer Math Adventure

W6
QUIZ

Week 6 Quiz | Exponents

Name: _____ Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

Section 1 – Evaluate (1 pt each)

1. $4^3 =$

2. $2^7 =$

3. $9^0 =$

4. $10^5 =$

5. $\sqrt{64} =$

6. $\sqrt{121} =$

Section 2 – Order of Operations (2 pts each)

7. $2^4 + 6 \times 3 =$

8. $(5 - 2)^3 \div 9 =$

9. $5^2 - 3^2 + 4 =$

10. $100 \div 10^2 \times 3 =$

Section 3 – Compare & Reason (2 pts each)

11. Circle the greater value: 3^5 or 5^3 Show your work below.

12. A square room has an area of 196 sq ft. What is the length of each wall? Show work.

Name: _____

Date: _____

EXPRESSIONS

Variable: A letter that stands for an unknown number. (x, n, a, b...)

Expression: Numbers, variables, and operations — but NO equals sign. ($3x + 5$)

Term: Parts of an expression separated by + or -. In $3x + 5$: terms are $3x$ and 5

Coefficient: The number multiplied by a variable. In $3x$, the coefficient is 3 .

Part A – Vocabulary Match

Label each part of the expression: $4n - 7 + 2n$

Expression: $4n - 7 + 2n$

Number of terms:

List all terms:

Coefficient of the first term:

Constant (number with no variable):

Part B – Write as an Algebraic Expression

Let n = the unknown number.

1. A number plus 8 →

2. 5 times a number →

3. A number divided by 4 →

4. 12 minus a number →

5. Twice a number, plus 3 →

6. A number squared, minus 1 → Score: _____ / 12

Name: _____

Date: _____

EXPRESSIONS

Like Terms have the same variable (and exponent). You can combine them by adding/subtracting coefficients.

$3x + 5x = 8x$ | $7a - 2a = 5a$ | *BUT:* $3x + 5y$ cannot be combined — different variables!

Tip: Constants (plain numbers) are also like terms with each other: $4 + 9 = 13$

Part A – Identify Like Terms

Group the like terms by circling them with matching colors (or labels A, B, C...).

1. $3x$ $5y$ $2x$ 8 $4y$ 11 $7x$

Like-term groups:

2. $6a$ $2b$ $3a^2$ $4a$ $9b$ a^2

Like-term groups:

Part B – Combine Like Terms

3. $4x + 3x =$

4. $9y - 2y =$

5. $5a + 2a + 3a =$

6. $8m - 5m + m =$

7. $3x + 4 + 2x + 7 =$

8. $6n - 2 + n + 5 =$

9. $4a + 3b - a + 2b =$

10. $10x + 5 - 4x - 3 =$

Summer Math Adventure

W7
D3

Week 7 · Day 3 | Writing & Solving One-Step Equations

Name: _____

Date: _____

EQUATIONS

Equation = two expressions set equal. There IS an = sign. $3x + 5 = 20$

Solve = find the value of the variable that makes it true.

Strategy (inverse operations): Whatever is done to the variable, do the opposite to both sides.

$x + 6 = 14 \rightarrow$ subtract 6 from both sides $\rightarrow x = 8$

$3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$

Part A – Name the Inverse Operation

1. $x + 9 = 15 \rightarrow$ use:

2. $x - 4 = 11 \rightarrow$ use:

3. $5x = 35 \rightarrow$ use:

4. $x \div 3 = 8 \rightarrow$ use:

Part B – Solve. Show Your Check.

5. $x + 12 = 30$

$x =$

Check: $___ + 12 = 30 \checkmark$

6. $y - 7 = 19$

$y =$

Check: $___ - 7 = 19 \checkmark$

7. $4n = 48$

$n =$

Check: $___$

8. $\frac{m}{6} = 9$

$m =$

Check: $___$

9. $t + 25 = 100$

$t =$

10. $8p = 72$

$p =$

Summer Math Adventure

W7
D4

Week 7 · Day 4 | Two-Step Equations

Name: _____

Date: _____

EQUATIONS

Two steps: undo addition/subtraction FIRST, then multiplication/division.

Example: $2x + 3 = 11$

Step 1: Subtract 3 from both sides $\rightarrow 2x = 8$

Step 2: Divide both sides by 2 $\rightarrow x = 4$

Always check: $2(4) + 3 = 8 + 3 = 11 \checkmark$

Part A – Label the Steps

Solve $3n - 5 = 16$. Fill in the blanks:

Step 1: Add ____ to both sides $\rightarrow 3n = \underline{\hspace{2cm}}$

Step 2: Divide both sides by ____ $\rightarrow n = \underline{\hspace{2cm}}$

Check: $3(\underline{\hspace{1cm}}) - 5 = \underline{\hspace{1cm}} \checkmark$

Part B – Solve. Show All Steps & Check.

1. $2x + 7 = 21$

$x =$

2. $4y - 3 = 17$

$y =$

3. $5m + 10 = 45$

$m =$

4. $\frac{n}{3} + 4 = 10$

$n =$

5. $6a - 12 = 0$

6. $3t + 15 = 60$

Name: _____

Date: _____

EQUATIONS

Inequality uses $<$, $>$, $\leq$, $\geq$ instead of $=$. The answer is a range of values.

$x + 3 > 7 \rightarrow x > 4$ (any number greater than 4 works)

Graphing: Open circle $\circ$ for $< / >$ (not equal) · Closed circle $\bullet$ for $\leq / \geq$ (includes the value)

Arrow points in the direction of solutions.

Part A – Write the Inequality

1. "x is greater than 5" $\rightarrow$

2. "y is at most 12" $\rightarrow$

3. "n is less than or equal to 8" $\rightarrow$

4. "t is at least 20" $\rightarrow$

Part B – Solve the Inequality

5. $x + 4 > 9 \rightarrow$

6. $y - 3 \leq 7 \rightarrow$

7. $2n < 16 \rightarrow$

8. $\frac{m}{4} \geq 3 \rightarrow$

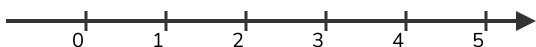
9. $3x - 1 > 11 \rightarrow$

10. $5 + 2y \leq 19 \rightarrow$

Part C – Graph the Solution on the Number Line

Draw an open or closed circle, then shade the arrow.

11. $x > 3$



Summer Math Adventure

W7
QUIZ

Week 7 Quiz | Expressions & Equations

Name: _____ Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

Section 1 – Expressions (1 pt each)

1. Write an expression: "6 less than 3 times x"

2. Evaluate $4n - 2$ when $n = 7$ =

3. Simplify: $5x + 3 + 2x - 1 =$

4. Simplify: $6a - 2b + a + 3b =$

Section 2 – Equations (2 pts each)

5. $x + 17 = 40$

x =

6. $7y = 63$

y =

7. $3m + 8 = 29$

m =

8. $\frac{n}{4} - 2 = 5$

n =

Section 3 – Word Problem & Inequality (2 pts each)

9. A box of crayons costs \$5. Lena also buys a notebook for \$3. She spends \$23 total. How many boxes of crayons did she buy?

Equation:

Week 7 Quiz – Expressions & Equations

Total: _____ / 20

Name: _____ Date: _____

PROBABILITY

Probability = how likely an event is to happen.

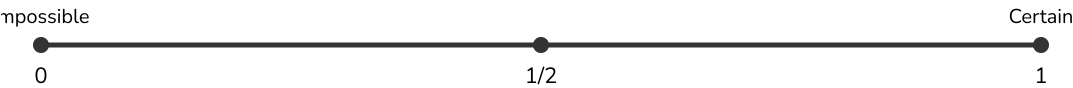
$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$

Scale: 0 (impossible) ←————→ 1 (certain)

$P = 0.5$ or $\frac{1}{2}$ means equally likely as unlikely.

Part A – Probability Scale

Place each event on the probability scale below. Write the letter at the right spot.



- A. Rolling a 7 on a standard die
- B. Flipping heads on a fair coin
- C. The sun rises tomorrow
- D. Picking a red card from a standard deck
- E. It will snow in Arizona in July
- F. Rolling an even number on a die

Part B – Calculate the Probability

A bag has 3 red, 4 blue, and 5 green marbles. Write each probability as a fraction.

1. $P(\text{red}) =$

2. $P(\text{blue}) =$

3. $P(\text{green}) =$

4. $P(\text{NOT red}) =$

5. $P(\text{red or blue}) =$

6. $P(\text{purple}) =$

Part C – Probability as Decimal and Percent

Use the same bag (3 red, 4 blue, 5 green marbles). Express each probability as a fraction, decimal, and percent.

7. $P(\text{green})$:

Fraction: _____
8. $P(\text{blue})$:

Fraction: _____
- Score: _____ / 14 + scale + table

Name: _____ Date: _____

PROBABILITY

Theoretical Probability: What SHOULD happen based on math. $P(\text{heads}) = \frac{1}{2}$

Experimental Probability: What ACTUALLY happens when you do the experiment.

If you flip a coin 20 times and get 12 heads $\rightarrow P(\text{heads}) = \frac{12}{20} = \frac{3}{5}$

The more trials you run, the closer experimental gets to theoretical.

Part A – Coin Flip Experiment Data

Omar flipped a coin 30 times. Results are below.

Outcome	Tally	Frequency
Heads		
Tails		

1. Experimental $P(\text{Heads}) =$

2. Experimental $P(\text{Tails}) =$

3. Theoretical $P(\text{Heads}) =$

4. Are they the same? Why/why not?

Part B – Die Roll Experiment

A die was rolled 60 times. Fill in the theoretical, then compare.

Number	1	2	3	4	5	6
Experimental Freq.	12	8	11	9	13	7
Experimental Prob.						
Theoretical Prob.						

5. Which number had experimental probability closest to theoretical?

6. If you rolled 600 times instead of 60, would you expect experimental to be closer to or farther from theoretical? Explain.

Name: _____

Date: _____

PROBABILITY

Sample Space: All possible outcomes of an experiment.

Tree Diagram: A visual tool to list all outcomes systematically.

Example – Flip a coin AND roll a die:

H → 1, 2, 3, 4, 5, 6 T → 1, 2, 3, 4, 5, 6 Total: 12 outcomes

Counting Principle: # of outcomes = options × options × ...

Part A – List the Sample Space

1. Flip a coin (H/T) and spin a spinner with colors Red, Blue, Green.

Complete the sample space:

{ ____, ____, ____, ____, ____, ____ }

Total outcomes: ____

Part B – Draw a Tree Diagram

2. A lunch special has 2 sandwich choices (Turkey, Veggie) and 3 drink choices (Water, Juice, Milk).

Draw a tree diagram and list ALL combinations.



Total combinations: ____

Part C – Use the Counting Principle

3. A shirt comes in 4 colors and 3 sizes. How many combinations?

4. You roll 2 dice. How many total outcomes?

5. A password uses 1 letter (A–E) and 1 digit (1–5). How many passwords?

6. You flip 3 coins. How many total outcomes?

Part D – Probability from a Sample Space

Week 8 · Day 3 – Sample Space & Tree Diagrams

Score: _____ / 10

Use the Turkey/Veggie × Water/Juice/Milk sample space from Part B.

7. $P(\text{Turkey AND Juice}) =$

8. $P(\text{any Veggie combo}) =$

Summer Math Adventure

W8
D4

Week 8 · Day 4 | Making Predictions with Probability

Name: _____

Date: _____

PROBABILITY

Prediction Formula: Expected outcomes = $P(\text{event}) \times \text{number of trials}$

Example: $P(\text{rolling a 3}) = \frac{1}{6}$. If you roll 60 times: expected 3s = $\frac{1}{6} \times 60 = 10$

This is a *prediction*, not a guarantee — actual results may vary!

Part A – Calculate Expected Outcomes

A spinner has 5 equal sections: 1 Red, 1 Blue, 2 Yellow, 1 Green.

1. $P(\text{Red}) =$

2. $P(\text{Yellow}) =$

3. If you spin 50 times, expected # of Red:

4. If you spin 50 times, expected # of Yellow:

5. If you spin 100 times, expected # of Green:

6. If you spin 30 times, expected # of Blue:

Part B – Real-World Predictions

7. A factory knows 3 out of every 200 light bulbs are defective. If they produce 1,000 bulbs, how many do they expect to be defective?

Answer:

8. A survey of 50 students showed 20 prefer math class. If there are 350 students in the school, how many would you predict prefer math?

Answer:

Summer Math Adventure

W8
D5

Week 8 · Day 5 | Probability Review & Mixed Practice

Name: _____

Date: _____

PROBABILITY

Week 8 Summary:

- ✓ $P(\text{event}) = \text{favorable} \div \text{total}$ | ✓ 0 = impossible, 1 = certain
- ✓ Theoretical vs. Experimental | ✓ Tree diagrams = list all outcomes
- ✓ Predictions = $P \times \text{trials}$ | ✓ Complement: $P(\text{NOT event}) = 1 - P(\text{event})$

Part A – Mixed Probability Problems

A standard deck has 52 cards: 4 suits (♠ ♥ ♦ ♣), 13 cards each, including Ace, 2–10, Jack, Queen, King.

1. $P(\text{drawing a heart}) =$

2. $P(\text{drawing a King}) =$

3. $P(\text{drawing a red card}) =$

4. $P(\text{NOT a spade}) =$

5. $P(\text{a Jack or Queen}) =$

6. $P(\text{a number card 2–9 of clubs}) =$

Part B – Complement Rule

$P(\text{NOT event}) = 1 - P(\text{event})$

7. $P(\text{rain}) = 0.3$. $P(\text{no rain}) =$

8. $P(\text{win}) = \frac{2}{5}$. $P(\text{not win}) =$

9. $P(\text{red marble}) = \frac{3}{8}$. $P(\text{not red}) =$

10. $P(\text{NOT rolling a 6}) =$

Part C – Scenario & Decision

11. A raffle sells 200 tickets. You buy 5. What is the probability you win?

$P(\text{win}) =$

Week 8 · Day 5 – Probability Review

Score: _____ / 12

Summer Math Adventure

W8
QUIZ

Week 8 Quiz | Probability

Name: _____ Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

Section 1 – Probability Calculations (1 pt each)

A jar has 6 red, 3 blue, and 1 yellow marble (10 total).

1. P(red) =

2. P(blue) =

3. P(yellow) =

4. P(NOT red) =

5. P(red or blue) =

6. P(green) =

Section 2 – Tree Diagram & Counting (2 pts each)

7. You choose a flavor (Chocolate, Vanilla) and a cone (Waffle, Sugar, Cup). Draw a tree diagram and list all combinations.

Total: ____

8. From that same sample space, what is P(Chocolate AND Waffle cone)?

P =

Section 3 – Prediction & Reasoning (2 pts each)

9. $P(\text{landing on blue}) = \frac{1}{4}$. If you spin 80 times, how many blues do you expect?

Answer:

Answer Key – Weeks 6, 7 & 8

For parent/teacher use only. Cut or fold before giving booklet to student.

WEEK 6 – EXPONENTS

Day 1 – What Is an Exponent?

Part A:

- base=2, exp=5, "two to the fifth power"
- base=10, exp=3, "ten cubed"
- base=6, exp=2, "six squared"
- base=4, exp=0, value=1

Part B:

- $3 \times 3 \times 3 = 27$
- 64
- 25
- 10,000
- 1
- 64

Part C:

- 7^3
- 9^2
- 2^5
- 10^3

Challenge: $2^8=256$, $4^4=256$ — equal!

Day 2 – Powers of 10

Table: 100; $10 \times 10 \times 10 = 1,000$; 10,000; 100,000; 1,000,000

Part B:

- 700
- 30,000
- 90
- 600,000
- 25,000
- 1,400

Part C: 7. 6 8. 5 9. 4 10. 2

Part D: 11. 300 12. 200,000 13. 300,000,000

Day 3 – Comparing Expressions

Part A:

- $81 > 64$ so $>$
- $128 > 49$ so $>$
- $125 < 243$ so $<$
- $36 < 64$ so $<$
- $=$
- $=$

Part B:

- 8, 9, 16, 4 $\rightarrow$ 4, 8, 9, 16
- 1, 32, 27, 100 $\rightarrow$ 1, 27, 32, 100

Part C:

- F (=8)
- F (=1)
- T
- F (=1)

Day 4 – PEMDAS with Exponents

Part A:

- $5+8=13$
- $16-10=6$
- $3 \times 16=48$
- $(4)^2=16$
- $8+9=17$
- $50-12=38$

Part B:

- $25-12=13$
- $36 \div 4 + 5 = 14$
- $27-18=9$

Part C:

- $(3+5)^2 \times 2 = 128$
- $4 \times (2+1)^2 = 36 \neq 9$; try $4 \times 2 + 1 = 9$ ✓

Day 5 – Perfect Squares & Square Roots

Chart:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Part B: 1.6 2.9 3.12 4.7

5.11 6.1

Part C: 7.No 8.Yes,8

9.Yes,10 10.No

Part D: 11. 8m 12. 8 rows

13. 81 cm^2

Challenge: $7^2=49$, $8^2=64$, so $7 < \sqrt{50} < 8$

Week 6 Quiz

Sec 1: 1.64 2.128 3.1 4.100,000 5.8 6.11

Sec 2: 7. $16+18=34$ 8. $27 \div 9=3$ 9. $25-9+4=20$ 10. $100 \div 100 \times 3=3$

Sec 3: 11. $3^5=243 > 5^3=125$; circle 3⁵ 12. $\sqrt{196}=14$; each wall = 14 ft

WEEK 7 – EXPRESSIONS & EQUATIONS